Luke Caffarey

Raleigh, NC | 919-695-6651 | lacaffar@ncsu.edu
linkedin.com/in/luke-caffarey-212473322 | github.com/lacaffar | lacaffar.github.io

Mechanical engineering student (GPA 3.889) with hands-on experience across fusion research, utility-scale wind, and cryogenic hardware design. Focused on building reliable hardware in service of SpaceX's mission to make life multiplanetary.

EDUCATION

North Carolina State University *Raleigh, NC*

B.S. Mechanical Engineering | Expected May 2028 | GPA: 3.889

- Relevant tools and coursework: SolidWorks, MATLAB, Python, thermodynamics, fluid mechanics, mechanics of materials.

EXPERIENCE

GE Vernova, Wind Engineer Intern *May 2026 to Present | Greenville, SC*

- Support engineering on utility-scale wind turbine hardware within GE Vernova's onshore wind business.
- Apply CAD and mechanical analysis to hardware that must operate reliably in the field for decades.

FPAC Laboratory, NC State, Undergraduate Researcher *Dec 2024 to Apr 2025 | Raleigh, NC*

- Analyzed spectroscopy and video data from KSTAR fusion reactor experiments to support reactor wall stability.
- Built Python workflows for time-series signal processing, using adaptive Gaussian smoothing to improve peak detection.

PROJECTS

Cryogenic Rotary Union, Independent Hardware Project *2026*

- Designing a sub-scale, bench-top cryogenic rotary union to transfer liquid nitrogen (LN2, -196 C) across a rotating joint, demonstrating extreme thermal gradient, dynamic sealing, and first-principles thermal-structural design.
- Architecture: 316L stainless inner fluid shaft, vacuum-jacketed and Aerogel insulation, spring-energized PTFE lip seals, and dry ceramic hybrid bearings thermally isolated from the cold core.
- Executing full CAD, coupled thermal-structural FEA, and fluid CFD, followed by machining, RTD and pressure-transducer instrumentation, and live rotational testing.

RCA Semantic Search Tool, Software Project

- Built a semantic search tool over root-cause-analysis records to retrieve similar past failures and resolutions.

LEADERSHIP AND INVOLVEMENT

Windpack Club, NC State, President *Aug 2025 to Present*

- Founded a 10-member team to compete in the U.S. Department of Energy Collegiate Wind Competition.
- Lead design, fabrication, and testing of scaled wind turbine systems and sustainability proposals.

Student Energy Club, NC State, Treasurer *Aug 2025 to Present*

- Registered the club as a 501(c)(3) non-profit organization.
- Initiated sponsor outreach, establishing connections with 5+ potential industry partners.

ADDITIONAL EXPERIENCE

Annelore's German Bakery, Assistant *Aug 2025 to Present | Cary, NC*

- Support weekend operations (16 hrs/week): baking assistance, order taking, and food prep.

Wake County Government, EMS Division, EMS Cadet *Jan 2022 to May 2024 | Raleigh, NC*

- Shadowed paramedics in 911 emergency response, assisting with patient care in high-pressure conditions.

SKILLS

CAD and Design: SolidWorks, GD&T;, design for manufacturing
Analysis: Thermal-structural FEA, fluid CFD, MATLAB, Python (signal processing, data analysis)
Hardware: Machining, fabrication, sensor instrumentation (RTDs, pressure transducers), cryogenics
Materials and Mechanics: 316L stainless, PTFE seals, ceramic bearings, thermodynamics, fluid mechanics